1 How to setup Camera Module 3 on Raspberry Pi 5's Ubuntu 24.04 ROS2 Jazzy system.

1. Connect Camera Module 3 to Camera Port 0.
2. Install Ubuntu 24.04 system to memory card.
3. Boot the Ubuntu 24.04 and complete basic configuration.
4. Install ROS2 Jazzy (follow official installation guide for Ubuntu deb)
5. Edit /boot/firmware/config.txt and make the following changes.
    a. Modify the camera_auto_detect setting to 0
    b. Add "dtoverlay=imx708,cam0"
6. Save the config.txt and reboot the system.
7. Install python3-ply

$ sudo apt install python3-ply

8. Create ROS2 workspace and build the camera_ros package from source code. Follow the official guide to build from source code.
    a. https://docs.ros.org/en/rolling/p/camera_ros/__README.html
9. Steps

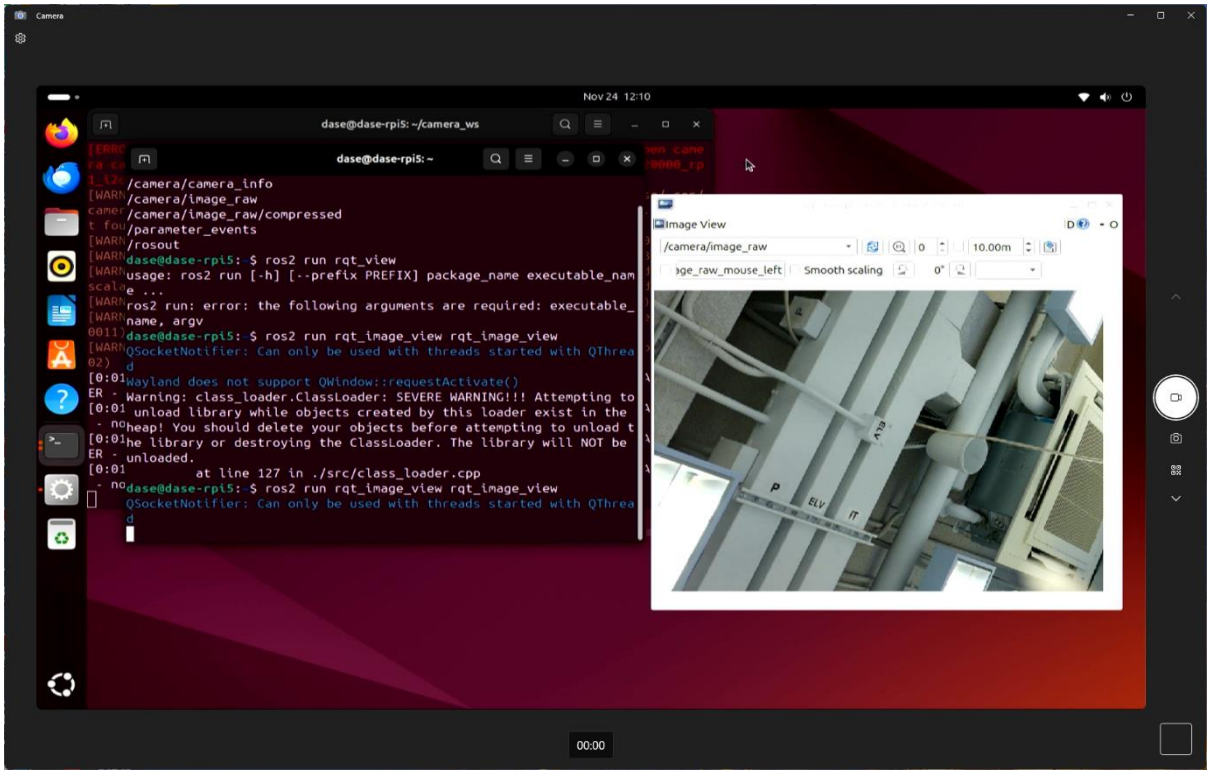
```
# create workspace
$ mkdir -p ~/camera_ws/src
$ cd ~/camera_ws/src

# check out libcamera
$ sudo apt -y install python3-colcon-meson
# Option A: official upstream
# git clone https://git.libcamera.org/libcamera/libcamera.git
# Option B: raspberrypi fork with support for newer camera modules
$ git clone https://github.com/raspberrypi/libcamera.git

# check out this camera_ros repository
$ git clone https://github.com/christianrauch/camera_ros.git

# resolve binary dependencies and build workspace
$ source /opt/ros/$ROS_DISTRO/setup.bash
$ cd ~/camera_ws/
$ rosdep install -y --from-paths src --ignore-src --rosdistro $ROS_DISTRO -
-skip-keys=libcamera
$ colcon build --event-handlers=console_direct+
```

10. If build succeed, launch a camera_ros node for testing.

$ ros2 run camera_ros camera_node

11. View camera output by rqt_image_view

$ ros2 run rqt_image_view rqt_image_view


Camera
Nov 24 12:10
dase@dase-rpi5: ~/camera_ws
dase@dase-rpi5: ~
/camera/camera_info
/camera/image_raw
/camera/image_raw/compressed
/parameter_events
/rosout
dase@dase-rpi5: $ ros2 run rqt_view
usage: ros2 run [-h] [--prefix PREFIX] package_name executable_name ...
ros2 run: error: the following arguments are required: executable_name, argv
dase@dase-rpi5: $ ros2 run rqt_image_view rqt_image_view
QSocketNotifier: Can only be used with threads started with QThread
Wayland does not support QWindow::requestActivate()
Warning: class_loader.ClassLoader: SEVERE WARNING!!! Attempting to unload library while objects created by this loader exist in the heap! You should delete your objects before attempting to unload the library or destroying the ClassLoader. The library will NOT be unloaded.
at line 127 in ./src/class_loader.cpp
dase@dase-rpi5: $ ros2 run rqt_image_view rqt_image_view
QSocketNotifier: Can only be used with threads started with QThread
Image View
/camera/image_raw
10.00m
Smooth scaling
00:00

12. Done